

JIARUI (RAY) FANG

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EDUCATION

Doctor of Philosophy in Psychology, Boston University

08/2024–Expected 05/2029

- ♦ Brain, Behavior, and Cognition Program
- ♦ Advisor: Prof. Michael Hasselmo
- ♦ GPA: 3.9 / 4.0

Master of Arts, Psychology, Boston University

05/2026

Bachelor of Science, Cornell University

08/2020-05/2024

- ♦ Double Majors: Biological Sciences / Biometry & Statistics
- ♦ GPA: 4.2 / 4.3; Summa Cum Laude; Distinction in Research

RESEARCH EXPERIENCE

Graduate Research Fellow

08/2024–Present

Computational Neurophysiology Laboratory with Prof. Michael Hasselmo, Boston University

Hippocampal Acetylcholine Dynamics During Spatial Exploration

- ♦ Characterized dorsal CA1 acetylcholine dynamics during free exploration using GRAB-ACh3.0 fiber photometry and synchronized behavioral tracking.
- ♦ Integrated dual-wavelength photometry, TTL-synchronized video, and DeepLabCut tracking to align cholinergic signals with frame-resolved locomotor behavior.
- ♦ Used linear mixed-effects models and cross-validation to show that CA1 acetylcholine activity is better explained by log-transformed locomotor speed than by raw speed.
- ♦ Extended the movement-based model to object- and location-novelty paradigms.

Computational Modeling of Egocentric Boundary Coding

- ♦ Developed and benchmarked receptive-field models for 174 retrosplenial cortex egocentric boundary cells from two published electrophysiology datasets.
- ♦ Implemented a ray-tracing framework to model neural responses to multiple visible boundary segments relative to the animal's position and heading.
- ♦ Compared angle–distance, isotropic, elliptical, and related receptive-field formulations using information criterion.
- ♦ Extended the model framework to allow negative response gain, identifying candidate inverse-EBC response profiles.

Undergraduate Research Assistant

06/2023-08/2023

Churchland Lab with Prof. Anne Churchland, UCLA

- ♦ Selected for the UCLA Neuroscience Scholars Program.
- ♦ Recorded medial prefrontal cortical spiking activity with Neuropixels in rats performing a multisensory decision-making task.
- ♦ Used Keypoint-MoSeq to quantify subtrial postural dynamics from behavioral videos.

Undergraduate Research Assistant

09/2022-05/2024

Computational Physiology Lab with Prof. Thomas Cleland, Cornell University

- ♦ Utilized transfer learning techniques to enhance the functionality of the open-source computer vision package, Annolid, enabling the detection of digging behavior in mice.
- ♦ Developed and validated an automated pipeline for scoring mouse digging behavior, achieving an AUROC of 0.98.

Undergraduate Research Assistant

09/2021-05/2024

The Laboratory of Neurobiology of Learning and Memory with Prof. David Smith, Cornell University

- ♦ Conducted a context-dependent odor discrimination task in rats to investigate neural correlates of odor memory.
- ♦ Recorded anterior olfactory nucleus activity with tetrodes during a context-dependent odor memory task and identified preliminary context-specific response patterns.

High School Research Assistant

11/2016-06/2019

Prof. Yi Zhong, Tsinghua University

- ♦ Utilized trans-Tango technique for downstream neuron staining.

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TEACHING EXPERIENCE

Teaching Fellow, PS 211 Introduction to Experimental Design

Spring 2026

- ♦ Led discussion sections, graded assignments and exams, held office hours, and provided individualized academic support.

Teaching Fellow, PS 337 Memory Systems

Fall 2025

- ♦ Led discussion sections on memory systems and behavioral neuroscience, prepared assessment materials, and held office hours.

Course Staff, CS 4780 Intro to Machine Learning

Fall 2023

- ♦ Updated course projects, coordinated grading sessions, held office hours, and supported course administration.

Course Staff, CS 2110 Object-Oriented Programming and Data Structures

Fall 2022

- ♦ Graded assignments and exams, held office hours, and answered questions on Ed Discussion for this Java course.

PUBLICATIONS

Peer reviewed:

Robinson, J. C., LaChance, P. A., Malmberg, S., Fang, J., Sankaranarayanan, S., Stern, C. E., & Hasselmo, M. E. (2026).

Trajectory scanning as a predictive coding mechanism for goal-directed navigation, obstacle avoidance and episodic memory. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 381(1954), 20250251.

<https://doi.org/10.1098/rstb.2025.0251>

Preprints:

Fang, J., LaChance, P. A., & Hasselmo, M. E. (2026). **Computational Structure of Egocentric Boundary Cell Responses in Retrosplenial Cortex.** SSRN. <https://doi.org/10.2139/ssrn.6427258>

Robinson, J. C., LaChance, P. A., Malmberg, S. L., Patel, M., Gross, E., Everett, D. E., Sankaranarayanan, S., Fang, J., & Hasselmo, M. E. (2025). **How does the response of egocentric boundary cells depend upon the coordinate system of environmental features?** bioRxiv. <https://doi.org/10.1101/2025.02.03.636060>

CONFERENCE ABSTRACTS AND PRESENTATIONS

Fang, J., Wilmot, J. H., & Hasselmo, M. E. (2026). **Logarithmic scaling of acetylcholine dynamics with movement speed in hippocampal CA1.** Poster presentation, International Symposium on Cholinergic Mechanisms, Poster 12.

Fang, J., LaChance, P. A., & Hasselmo, M. E. (2025). **Internal Representations Underlying Egocentric Boundary Cell Responses in Retrosplenial Cortex.** Poster presentation, Society for Neuroscience, Poster 305.12.

Fang, J., Yang, C., & Cleland, T. A. (2023). **Scoring rodent digging behavior with Annolid.** Poster presentation, Society for Neuroscience, Poster 512.01.

Wu, W.-Y., Turker, H., Fang, J., Cleland, T. A., Linster, C., & Smith, D. M. (2023). **Neuronal Responses in the Anterior Olfactory Nucleus During a Complex Odor Memory Task.** Poster presentation, Society for Neuroscience, Poster 238.13.

ADDITIONAL INFORMATION

Programming & Analysis: Python, R, MATLAB, Java, LaTeX, HTML; DeepLabCut, Keypoint-MoSeq, Annolid

Experimental Methods: Mouse and rat behavioral training, fiber photometry, electrophysiology, Drosophila behavioral assays

Languages: Mandarin Chinese (native), English (fluent)